```
import spacy
from spacy import displacy
from transformers import pipeline, AutoTokenizer, AutoModelForTokenClassification
import torch
import pandas as pd
```

```
!python -m spacy download en_core_web_sm
nlp = spacy.load("en_core_web_sm")
```

```
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 86.3 MB/s eta 0:00:00
✔ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
sentences = [
    "Apple announced a new iPhone in California.",
    "Lionel Messi scored two goals for Argentina.",
    "The Prime Minister of India visited Hyderabad.",
    "Microsoft released new AI tools in 2026.",
    "The Olympics will be held in Paris."
]
```

```
for sent in sentences:
    doc = nlp(sent)
    for ent in doc.ents:
        print(f"Sentence: {sent}\nEntity: {ent.text}\tLabel: {ent.label_}\n")
```

```
Sentence: Apple announced a new iPhone in California.
Entity: Apple	Label: ORG

Sentence: Apple announced a new iPhone in California.
Entity: iPhone	Label: ORG

Sentence: Apple announced a new iPhone in California.
Entity: California	Label: GPE

Sentence: Lionel Messi scored two goals for Argentina.
Entity: Lionel Messi	Label: PERSON

Sentence: Lionel Messi scored two goals for Argentina.
Entity: two	Label: CARDINAL

Sentence: Lionel Messi scored two goals for Argentina.
Entity: Argentina	Label: GPE

Sentence: The Prime Minister of India visited Hyderabad.
Entity: India	Label: GPE

Sentence: The Prime Minister of India visited Hyderabad.
Entity: Hyderabad	Label: ORG

Sentence: Microsoft released new AI tools in 2026.
Entity: Microsoft	Label: ORG

Sentence: Microsoft released new AI tools in 2026.
Entity: AI	Label: GPE

Sentence: Microsoft released new AI tools in 2026.
Entity: 2026	Label: DATE

Sentence: The Olympics will be held in Paris.
Entity: Olympics	Label: EVENT

Sentence: The Olympics will be held in Paris.
Entity: Paris	Label: GPE
```

```
results = []
for sent in sentences:
    doc = nlp(sent)
    for ent in doc.ents:
```

```
        results.append([sent, ent.text, ent.label_])

df_spacy = pd.DataFrame(results, columns=["Sentence", "Entity", "Label"])
df_spacy
```

	Sentence	Entity	Label
0	Apple announced a new iPhone in California.	Apple	ORG
1	Apple announced a new iPhone in California.	iPhone	ORG
2	Apple announced a new iPhone in California.	California	GPE
3	Lionel Messi scored two goals for Argentina.	Lionel Messi	PERSON
4	Lionel Messi scored two goals for Argentina.	two	CARDINAL
5	Lionel Messi scored two goals for Argentina.	Argentina	GPE
6	The Prime Minister of India visited Hyderabad.	India	GPE
7	The Prime Minister of India visited Hyderabad.	Hyderabad	ORG
8	Microsoft released new AI tools in 2026.	Microsoft	ORG
9	Microsoft released new AI tools in 2026.	AI	GPE
10	Microsoft released new AI tools in 2026.	2026	DATE
11	The Olympics will be held in Paris.	Olympics	EVENT
12	The Olympics will be held in Paris.	Paris	GPE

Next steps: Generate code with df_spacy New interactive sheet

```
displacy.render(doc, style="ent", jupyter=True)
```

The Olympics **EVENT** will be held in Paris **GPE** .

```
tokenizer = AutoTokenizer.from_pretrained("dslim/bert-base-NER")
model = AutoModelForTokenClassification.from_pretrained("dslim/bert-base-NER")
ner_pipeline = pipeline("ner", model=model, tokenizer=tokenizer)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

config.json: 100% 829/829 [00:00<00:00, 14.9kB/s]

tokenizer_config.json: 100% 59.0/59.0 [00:00<00:00, 1.23kB/s]

vocab.txt: 213k/? [00:00<00:00, 1.59MB/s]

added_tokens.json: 100% 2.00/2.00 [00:00<00:00, 41.6B/s]

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
```

special_tokens_map.json: 100% 112/112 [00:00<00:00, 1.84kB/s]

model.safetensors: 100% 433M/433M [00:06<00:00, 75.5MB/s]

Loading weights: 100% 199/199 [00:00<00:00, 585.88it/s, Materializing param=classifier.weight]

```
BertForTokenClassification LOAD REPORT from: dslim/bert-base-NER
Key                       | Status     |  |
--------------------------+------------+--+-
bert.pooler.dense.bias    | UNEXPECTED |  |
bert.pooler.dense.weight  | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

```
for sent in sentences:
    outputs = ner_pipeline(sent)
    print(f"Sentence: {sent}")
    for out in outputs:
        print(f"Word: {out['word']}\tEntity: {out['entity']}\tScore: {out['score']:.2f}")
    print()
```

```
Sentence: Apple announced a new iPhone in California.
Word: Apple     Entity: B-ORG   Score: 1.00
Word: iPhone    Entity: B-MISC  Score: 0.97
Word: California        Entity: B-LOC   Score: 1.00

Sentence: Lionel Messi scored two goals for Argentina.
Word: Lionel    Entity: B-PER   Score: 1.00
Word: Me        Entity: I-PER   Score: 1.00
Word: ##ssi     Entity: I-PER   Score: 1.00
Word: Argentina Entity: B-LOC   Score: 1.00

Sentence: The Prime Minister of India visited Hyderabad.
Word: India     Entity: B-LOC   Score: 1.00
Word: Hyderabad Entity: B-LOC   Score: 1.00

Sentence: Microsoft released new AI tools in 2026.
Word: Microsoft Entity: B-ORG   Score: 1.00
Word: AI        Entity: B-MISC  Score: 0.94

Sentence: The Olympics will be held in Paris.
Word: Olympics  Entity: B-MISC  Score: 0.99
Word: Paris     Entity: B-LOC   Score: 1.00
```

```
clean_results = []
for sent in sentences:
    outputs = ner_pipeline(sent)
    for out in outputs:
        word = out['word'].replace("##", "")  # remove subword markers
        clean_results.append([sent, word, out['entity'], round(out['score'], 2)])

df_hf = pd.DataFrame(clean_results, columns=["Sentence", "Entity", "Label", "Confidence Score"])
df_hf
```

	Sentence	Entity	Label	Confidence Score
0	Apple announced a new iPhone in California.	Apple	B-ORG	1.00
1	Apple announced a new iPhone in California.	iPhone	B-MISC	0.97
2	Apple announced a new iPhone in California.	California	B-LOC	1.00
3	Lionel Messi scored two goals for Argentina.	Lionel	B-PER	1.00
4	Lionel Messi scored two goals for Argentina.	Me	I-PER	1.00
5	Lionel Messi scored two goals for Argentina.	ssi	I-PER	1.00
6	Lionel Messi scored two goals for Argentina.	Argentina	B-LOC	1.00
7	The Prime Minister of India visited Hyderabad.	India	B-LOC	1.00
8	The Prime Minister of India visited Hyderabad.	Hyderabad	B-LOC	1.00
9	Microsoft released new AI tools in 2026.	Microsoft	B-ORG	1.00
10	Microsoft released new AI tools in 2026.	AI	B-MISC	0.94
11	The Olympics will be held in Paris.	Olympics	B-MISC	0.99
12	The Olympics will be held in Paris.	Paris	B-LOC	1.00

Next steps: Generate code with df_hf New interactive sheet

```
comparison = pd.DataFrame({
    "Feature": ["Model Type", "Speed", "Accuracy", "Context Handling", "Confidence Score"],
    "spaCy": ["Rule-based + statistical", "Fast", "Good but limited", "Weak in long context", "Not provided"],
    "Hugging Face": ["Transformer (BERT)", "Slower", "Higher F1", "Strong contextual", "Yes"]
})
comparison
```

	Feature	spaCy	Hugging Face
0	Model Type	Rule-based + statistical	Transformer (BERT)
1	Speed	Fast	Slower
2	Accuracy	Good but limited	Higher F1
3	Context Handling	Weak in long context	Strong contextual
4	Confidence Score	Not provided	Yes

Next steps: Generate code with comparison New interactive sheet

```
from datasets import load_dataset

# CoNLL-2003
# The 'datasets' library in its current version no longer supports loading datasets directly from scripts like 'conll2003.py'.
# Please consider alternative ways to load this dataset, such as specific revisions or direct file downloads, or use an alterna
# conll = load_dataset("conll2003")

# WNUT-17
# The 'datasets' library in its current version no longer supports loading datasets directly from scripts like 'wnut_17.py'.
# Please consider alternative ways to load this dataset, such as specific revisions or direct file downloads, or use an alterna
# wnut = load_dataset("wnut_17")
```